



3 Months Data Analytics **BOOTCAMP**



Contents

- Overview
- Program Delivery
- Curriculum
- Requirements
- Program
- Highlights





About us

Data-Lead Africa is a research and training consulting firm providing services in monitoring and evaluation, data analytics, data science training, and strategic consulting services. We deploy exceptional skills in statistics, research methods, and information technology in processing data into insights. We have our core expertise in research, data collection, data analysis training, data science training, monitoring and evaluation, business analytics, human resource analytics, and market research analytics.





As part of its commitment to developing strong data science capabilities, Data-Lead Africa has launched the Data Analytics Bootcamp.

The Data Analytics Bootcamp is a 3-month intensive program designed to train and equip aspiring data professionals with practical, job-ready skills in the field of Data Analytics. It is ideal for individuals who want to accelerate their careers and gain a competitive edge in today's data-driven world.

This program provides students with the tools, techniques, and practical knowledge needed to transform raw data into meaningful insights that support better decision-making within organizations.

Through hands-on projects and real-world applications, participants will learn how to apply their data analytics skills to solve industry-specific problems, preparing them to confidently contribute to the organizations they work for or aspire to join.

BOOTCAMP INFORMATION

Duration: 3 Months

Program Days: Monday – Friday

Program Fee : **NGN 250,000**

Location: Abuja (Physical) & Virtually

“Upgrade
your Data
Analytic



Career RoadMap



TOOLS YOU WILL MASTER AT THE END OF THE PROGRAM:



What we Offer

Our data analytics boot-camp offers hands-on training with real-world projects, exposure to industry standard tools, and mentorship from experienced professionals. Students gain practical skills in data analysis, visualization, and interpretation, equipping them for successful careers in data-driven world.

Hands-on Training

Students will be practically trained on different industry tools.

Mentorship

Students will be mentored by seasoned professionals and experts in different fields.

Self-Study

In addition to the trainings, Students will carry out personal research and self development in the course of the programme

In agency Internship

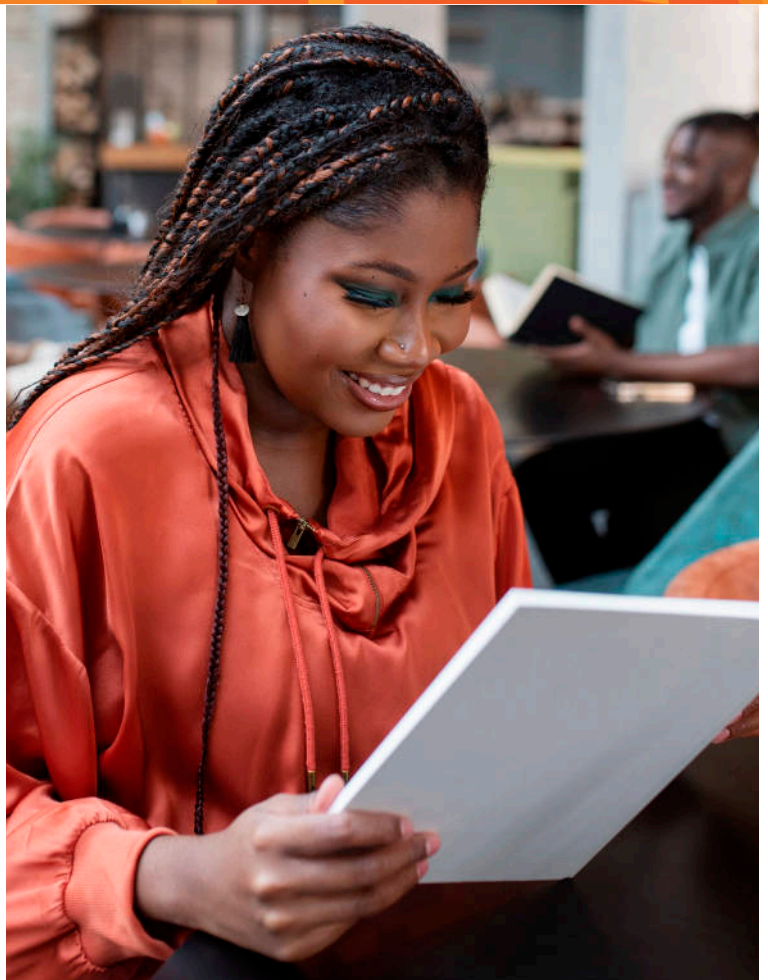
Students will have the opportunity to work alongside Data-Lead Africa team on projects and gain valuable work experience

Program Curriculum

Students for the Data Analytics programme will be required to complete

The core courses in:

- Data collection methods
- Data Analysis
- Data Management
- Data Visualization
- Capstone project





Data Collection

Overview

This programme is designed to make students understand the methods involved in data collection for public health research. At the end of the programme they should be familiar with the different data types, data collection techniques and be able to develop sample data collection tools for public health research.

Interns are required to take up courses which includes:

- Introduction to Data Collection
- Data Collection using KoboToolbox
- Form crating Using XLS

On-the-job work and learn experience.



Introduction to Data collection

This course is designed to help you gain direct insight into the ways data is collected, what techniques are appropriate for specific organizational needs, and how to conduct specific data collection based on the data collection goal and selected technique.

Syllables:

Here's what we will cover in this course

- Forms and Types of Data
- Method of Data collection
- DataValidation
- Data collection techniques
- Sources of data
- Introduction to Data collection tools

Mobile Data Collection Using Kobo Toolbox

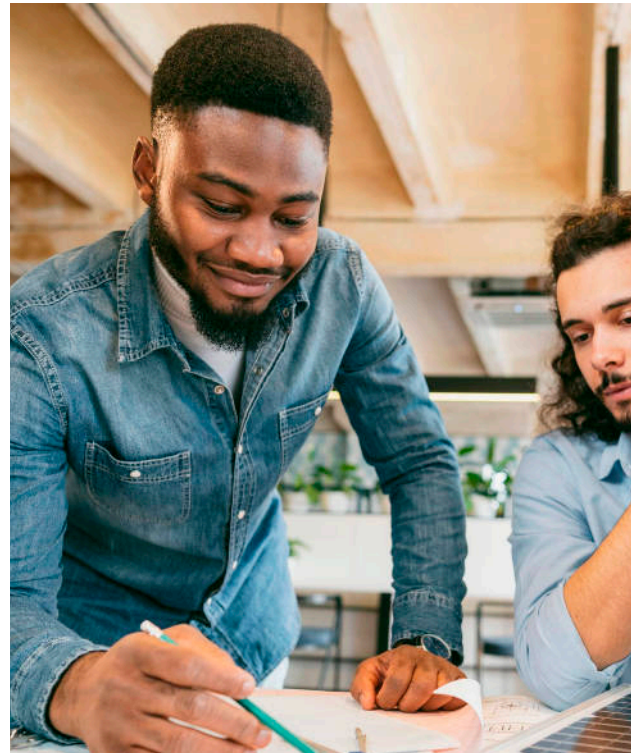
Technology has introduced new ways of collecting data for monitoring and evaluation that does not require the use of papers but through the Mobile Data Collection (Kobotoolbox).

Using Kobotoolbox, data is collected on the phone and transmitted to a server for analysis. This training will equip the participants with skills to author, field and manage mobile data collection platform using KoboToolbox

Syllabus:

Here is what we will cover in this course

- Introduction to Mobile Data Collection
- Introduction to Kobotoolbox
- Designing and Creating forms using Kobotoolbox
- Creating and Designing XLS form
- Kobotoolbox Data Collect
- Kobotoolbox Aggregate Server
- Visualizing Geographic Data



Data Analysis

Overview

Data Analysis is the lifeblood of today economy and many organizations both big and small now make decisions on data driven insights. This has led to a rise in demand for experienced data analysts. This course have been put together to equip students with core data skills which will be taught by a team of seasoned industry experts with data science, big data analytics expertise.

Students are required to take up courses which includes:

- Qualitative Data Analysis
- Quantitative Data Analysis
- Big Data Analysis

Qualitative Data Analysis

Overview

This course is designed to give us understanding of qualitative data and also the kind of analysis that can be done on qualitative data set. It will also introduce use the different industry tools used for qualitative data analysis.

Interns are required to take up courses which includes:

- Qualitative Data Analysis using QDA Miner
- Qualitative Data Analysis using NVIVO
- Qualitative Data Analysis using Atlas.Ti

Qualitative Data Analysis Using QDA Miner

QDA Miner is a data analysis software used by researchers in different field and sector to analyze qualitative data. This hands-on training will provide you with the information and practice you need to get started with your own project. Using sample data, you'll create a project in QDA Miner and will work with a range of material such as documents, PDFs, audio, video, pictures, spreadsheets and web data.

Here's what we'll cover in the course:

- Understanding Research design.
- Introduction and overview of QDA Miner.
- Starting a new Project in QDA Miner.
- Importing Document.
- Coding and Analysis in QDA Miner.
- Reporting.

Qualitative Data Analysis Using NVIVO

NVivo is a powerful qualitative data analysis software used by researchers in education, government, NGO and the commercial sectors worldwide. This hands-on training will provide you with the information and practice you need to get started with your own project. Using sample data, you'll create a project in NVivo and will work with a range of material such as documents, PDFs, audio, video, pictures, spreadsheets and web data.

Here's what we'll cover in the course:

- Understanding Qualitative data
- Introduction to NVIVO
- Data Management using NVIVO
- Coding in Nvivo
- Data Analysis
- Data Visualisation in Nvivo
- Reporting

Don't Quit



Quantitative Data Analysis

Overview

This course is designed to give us Understanding of quantitative data and also the kind of analysis that can be done on quantitative data set. It will also introduce use the different industry tools used for quantitative data analysis.

Interns are required to take up courses which includes:

- Quantitative Data Analysis using Excel
- Quantitative Data Analysis using SPSS
- Quantitative Data Analysis using STATA

Quantitative Data Analysis using MS. Excel

Overview

This course guides you through the fundamentals of using Microsoft Excel to carry out quantitative data analysis. Excel is typical

quantitative data analysis software and has so many tools you can use to perform various types of quantitative analysis.

Syllables:

Here's what we'll cover in the course:

- Introduction to statistical analysis
- Understanding quantitative data analysis
- Installation of excel data analysis ToolPak
- Descriptive analysis
- Inferential statistics
- Predictive analysis

Quantitative Data Analysis using Stata

Overview

This course guides you through the fundamentals of using Stata for typical data analysis process. Stata is an easy to use but very powerful data analysis software package. Stata offers a wide array of statistical tools that include both standard methods and newer, advanced methods, as new releases of Stata are distributed annually.

Syllables:

Here's what we'll cover in the course:

- Understanding Stata as a Data analysis software
- Navigating Stata Interface
- Exploring data on Stata
- Data management on Stata
- Basic statistical analysis on Stata
- Data visualization in Stata

Quantitative Data Analysis using SPSS

Overview

This course guides you through the fundamentals of using SPSS for typical data analysis process. Learn the basics of read data, data definition, data modification, and data analysis and presentation of your results.

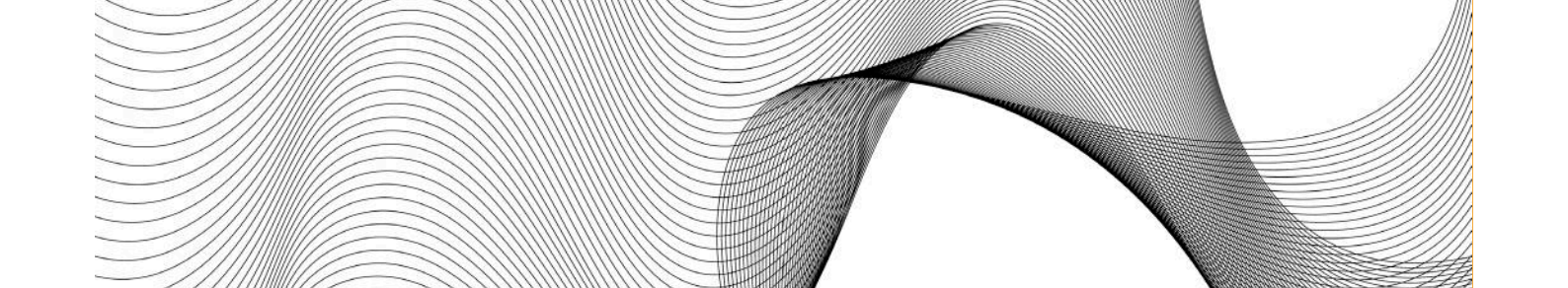
See how easy it is to get your data into SPSS so that you can focus on analyzing the information.

Syllables:

- Introduction to statistical analysis
- Understanding SPSS and its features
- Descriptive analytics using SPSS
- Inferential analytic using SPSS
- Predictive analytics using SPSS
- Data visualization in SPSS.

LEAD ➡ **With**
DATA-LEAD





Big Data Analysis

Big Data Analysis Using R

Overview

This is an introductory course on how to use the R programming language and software environment for data manipulations and munging, exploratory data analysis and data visualizations. Students will be familiar to the R ecosystem and learn how to use R for the most common data analysis tasks, including loading, cleaning, transforming, summarizing and visualizing

Syllables:

Here's what we'll cover in the course:

- Understanding to R Ecosystem
- Introduction to data types in R
- Basic Statistics in R
- Statistical Modelling
- Loops and Functions
- Data Visualization in R

Data Management Using SQL

Overview

Microsoft SQL Server is one the most popular Relational Database Management System (RDBMS) used in Microsoft universe. It can be used for data storage as well as for data retrieval for applications which can be either on desktop or Web.

Syllables:

Here's what we'll cover in the course:

- Introduction to SQL data science ecosystem Session
- Data wrangling and data exploring Session
- Data visualizations and time series analysis Session
- Practical case studies from finance and marketing

Geospatial Data Analysis

OVERVIEW

A Geographic Information System is a multi-component environment used to create, manage, visualize and analyze data and its spatial counterpart. By relating seemingly unrelated data, GIS can help individuals and organizations better understand spatial patterns and relationships.

SYLLABLES:

Here's what we'll cover in the course:

- Introducing GIS
- Vector Data
- Vector Attribute Data
- Data Capture
- Raster Data
- Topology
- Coordinate Reference Systems
- Map Production
- Vector Spatial Analysis (Buffers)
- Spatial Analysis (Interpolation)

Data Analysis

Overview

The data visualization Module is designed to help interns gain mastery in Data visualization. At the end of the course you will be able to master the process and methods of transforming raw data into insightful and captivating visuals using data visualization tools such as MS Excel, Power BI and Tableau Interns are required to take up courses which includes:

- Data Visualization using Excel
- Data Visualization using PowerBi
- Data Visualization using Tableau

Data Visualization Using Excel

Overview

Excel makes a good tool for data visualization: it is familiar and widely available, and it has excellent graphical capabilities. This course is designed to You will learn about the fundamentals of data visualization, how to create types of charts in Excel, and tips on how to get the most of the charts you make.

Syllables:

- Data visualization best practice
- Prepping your data for analysis
- Charting by types
- Dashboarding

What will I be able to do?

Work with your peers to hone your skills, and put them to use right away.

- Engage stakeholders through managerial and soft skills
- Build an M&E plan
- Design empirical project evaluation
- Evaluate impact using quantitative and qualitative approaches
- Develop a communication plan to share results, recommendations, and emerging practices

Data Visualization Using Power BI

Overview

Business intelligence (BI) combines business analytics, data mining, data visualization, data tools and infrastructure, and best practices to help organizations to make more data-driven decisions. In practice, you know you've got modern business intelligence when you have a comprehensive view of your organization's data and use that data to drive change, eliminate inefficiencies, and quickly adapt to market or supply changes

Syllables:

- Introduction to Power BI
- How to create dashboards,reports and visualizations
- How to use Power BI service (SaaS) and

Monitoring & Evaluation

Overview

Monitoring and evaluation (M&E) is an effective tool for enhancing the quality of project planning and management. Monitoring and evaluation specialists; project coordinators; programme managers; independent evaluators; donor staff appraising and evaluating projects and programmes; non-governmental organization officials involved in monitoring and evaluating

What topics does this course cover?

Identify areas of improvement and brush up on your technical and managerial skills.

- Results-based management
- Monitoring, reporting and control
- Risk management
- Evaluation of development programmes and projects
- Monitoring and evaluation system, methodologies, tools and technique

01

Module

KOBO TOOLBOX

Course Module

Module 01

Introduction to Mobile Data Collection and Kobo Toolbox

- Introduction to Data Collection
 - Overview of mobile data collection methods
 - Benefits and challenges of using mobile devices for data collection
 - Understanding the purpose and features of Kobo Toolbox
 - Creating an account and setting up a project
 - Overview of the Kobo Toolbox interface
-

Module 02

Understanding the Basics of Form Design and Deployment on Kobo Toolbox

- Principles of effective form design
 - Creating forms using the Kobo Form Builder
 - Adding different question types and logic to forms
 - Deploying forms to mobile devices
 - Testing forms on different devices and platforms
 - Managing form versions and updates
-

Module 03

Data Collection and Submission

- Conducting mock data collection exercises
 - Troubleshooting common data collection issues
 - Uploading collected data to the Kobo Toolbox server
 - Reviewing submitted data in the online interface
 - Managing data submissions and resolving discrepancies
-

Module 04

Data Analysis

- Exporting data from Kobo Toolbox for analysis
 - Using built-in visualization tools in Kobo Toolbox
 - Exporting data for visualization in external software (e.g., Excel, Tableau)
 - Creating simple visuals and reports from collected data
-

05

Advanced Form Design using XLS

- Advanced form design techniques (e.g., repeat groups, constraints)
- Building forms using XLS format
- Incorporating multimedia elements into forms
- Customizing the appearance and functionality of Kobo Toolbox forms and interface

02

Module

STATA CONTENT

Course Module

Module 01

Introduction to STATA

- Setting up and installation
 - Understanding the STATA interface
 - Creating a working environment
 - Creating a log file
 - Understanding how the do-file works
-

Module 02

Data management and exploration

- Importing data set
 - Data management
 - Generating variable
 - Data exploration
 - Variable management
 - Conversion of data types
-

Module 03

Understanding Descriptive Analysis using STATA

- Measures of count
 - Measure of dispersion
 - Interpretation of results generated
 - Measures of central tendency
 - Measure of distribution
 - Data reporting
-

Module 04

Exploratory Analysis

- Understanding Parametric and Non-Parametric assumptions
 - Understanding one-tail test and two-tail test
 - Understanding hypothesis testing
 - Understanding probability value, significance level, and confidence level
 - Decision-making process with inferential assumptions
-

Module 05

Fundamentals of Inferential Analysis

- Independent t-test
 - One-sampled t-test
 - Two-way ANOVA
 - Correlation and covariances
 - MANOVA
 - MANOVA
 - Paired t-test
 - One-way ANOVA
 - Chi-square
 - Posthoc test
 - ANCOVA
 - MANCOVA
-

Module 06

Predictive Analysis using STATA

- Understanding predictive analysis
- Linear regression models
- Simple linear regression
- Multiple linear regression
- Binary logistic regression
- Model prediction

03

Module R CONTENT

Course Module

Module 01

Introduction to R

- Setting up and installation
- Understanding the R interface
- Basic R codes
- Understanding data structures in R
- Storing variables

Module 02

Introduction to data exploration

- Creating a working directory
- Creating a new project
- R data exploration
- Understanding data types in R
- Installation of packages
- Performing basic operations
- Importing dataset

Module 03

Understanding descriptive analysis using R

- Measures of count
- Measure of dispersion
- Interpretation of results generated
- Measures of central tendency
- Measure of distribution
- Data reporting using R

Module 04

Exploratory Analysis on R

- Understanding Parametric and Non-Parametric assumptions
- Understanding one-tail test and two-tail test
- Understanding hypothesis testing
- Understanding probability value, significance level, and confidence level
- Decision-making process with inferential assumptions

Module 05

Fundamentals of Inferential Analysis using R

- Independent t-test
- One-sampled t-test
- Two-way ANOVA
- Correlation and covariances
- MANOVA
- MANOVA
- Paired t-test
- One-way ANOVA
- Chi-square
- Posthoc test
- ANCOVA
- MANCOVA

Module 06

Predictive Analysis

- Understanding predictive analysis
- Linear regression models
- Simple linear regression
- Multiple linear regression
- Binary logistic regression

04

Module NVIVO

Course Module

Module 01

Introduction to Qualitative Analysis and NVivo

- What is qualitative analysis
 - What is NVivo and how does it work?
 - Benefits of using NVivo for qualitative analysis
 - Setting up an NVivo project
-

Module 02

Importing Data

- Importing your dataset on Nvivo
 - Cleaning and organizing data for analysis
 - Understanding the text
-

Module 03

Coding and Analysis

- What is coding and how is it done in NVivo?
 - Creating codes and editing the codes
 - Analyzing and visualizing coded data
 - Use of Coding stripes and Highlights
 - Assigning codes to text
 - Exporting the Codebook
-

Module 04

Conclusion

- Best practices for qualitative analysis using NVivo
- Review of key concepts and skills learned
- Tips for using NVivo more effectively

05

Module QDA MINER

Course Module

Module 01

Introduction to QDA Miner

- Overview of the software
 - Applications of QDA Miner
 - How to Install QDA Miner
 - Creating a new project
 - Importing data from different sources
-

Module 02

Data Management using QDA Miner

- Importing various types of qualitative data (text, images, audio, video)
 - Data preparation and cleaning
 - Organizing data into projects and folders
 - Creating cases and variables on QDA miner
-

Module 03

Coding and Data Analysis

- Creating codes on QDA miner
 - Managing coding schemes and dictionaries
 - Introduction to coding (Manual coding vs. automated coding)
 - Assigning codes to data
 - Creating quotations
 - Combining codes
-

Module 04

Text Retrieval and Data Exploration on QDA miner

- Searching for coded segments
- Text retrieval techniques
- Exploring data through various visualizations (Word Clouds)
- Understanding thematic analysis
- Applying thematic analysis using QDA Miner
- Exporting reports (Word format & Excel Format)

06

Module POWER BI

Course Module

Module 01

Introduction to Power BI

- Overview of Power BI
 - Understanding Power BI Desktop vs. Power BI Service
 - Installation of Power BI Desktop
 - Understanding the interface of Power BI Desktop
 - Connecting to various data sources (Excel, databases, web sources)
 - Loading data into Power BI Desktop
-

Module 02

Data Transformation & Modeling

- Introduction to Power Query Editor
 - Data cleaning, shaping, and transformation techniques
 - Understanding data modeling concepts (relationships, tables, measures)
 - Creating relationships between tables
 - Implementing calculated columns and measures
-

Module 03

Data Visualization and Building Interactive Dashboards

- Creating basic visualizations (bar charts, line charts, pie charts)
 - Formatting and customizing visuals on Power BI
 - Building interactive reports using slicers, filters, and drill-through
 - Creating dashboards to visualize key insights from the data
-

Module 04

Advanced Data Modeling and DAX Measures

- Overview of Data Analysis Expressions (DAX)
 - Writing basic DAX expressions for calculations and measures
 - Time intelligence functions
 - Iterating functions and row context vs. filter context
-

Module 05

Advanced Visualization Techniques and Introduction to Power BI Service

- Custom visuals and marketplace visuals
 - Drill-down and drill-through options
 - Publishing reports and dashboards to Power BI Service
 - Collaboration and sharing options on Power BI Service
 - Configuring data refresh settings
 - Troubleshooting data refresh issues
-

07

Module MICROSOFT EXCEL

Course Module

Module 01

Introduction to Excel

- Overview of Excel and its features
- Understanding the Excel interface and navigation

Module 02

Basic Data Entry and Formatting

- Entering data into cells
- Formatting text, numbers, and cells

Module 03

Basic Formulas and Functions

- Introduction to basic formulas (SUM, AVERAGE, COUNT)
- Using functions for simple calculations

Module 04

Data Management Techniques

- Sorting and filtering data
- Data validation and error checking

Module 05

Advanced Formulas and Functions

- Working with logical functions (IF, AND, OR)
- Using lookup functions (VLOOKUP, HLOOKUP, INDEX/MATCH)

Module 06

PivotTables

- Introduction to PivotTables and Pivot Charts
- Creating PivotTables from data sets
- Analyzing data with PivotTables

Module 07

Excel Macros and VBA

- Introduction to Excel Macros
- Recording and editing macros
- Introduction to Visual Basic for Applications (VBA)

Module 08

Data Visualization: Charts and Graphs

- Creating different types of charts (bar charts, line charts, pie charts)
- Customizing chart elements and formatting

Module 09

Advanced Charting Techniques

- Creating combination charts and sparklines
- Using trendlines and error bars

08

Module

PYTHON FOR DATA ANALYSIS

Course Module

Module 01

Foundations of Data Management and Python Environment

- Conceptual Foundation
 - Introduction to Python
 - Understanding the Python Interface (JupyterLab)
 - Setting Up a Professional Workflow
 - Data Handling Basics
-

Module 02

Data Exploration and Core Data Management (Pandas)

- Exploring Data
 - Managing Variables (Columns)
 - Data Cleaning
 - Data Restructuring
-

Module 03

Descriptive Analysis and Interpretation

- Measures of Count
 - Measures of Central Tendency
 - Measures of Dispersion
 - Measures of Distribution
 - Interpretation of Results
-

Module 04

Foundations of Inferential Thinking

- Statistical Assumptions
 - Hypothesis Testing
-

Module 05

Core Inferential Analysis in Python (SciPy & Statsmodels)

- Mean Comparison Tests
 - Analysis of Variance
 - Association Tests
-

Module 06

Regression and Predictive Analysis (Statsmodels & Scikit-learn)

- Understanding Predictive Analysis
 - Linear Regression
 - Logistic Regression
 - Model Prediction
-

Module 07

Data Visualization in Python (Matplotlib & Seaborn)

- Basic Graphs
 - Interpreting Visual Outputs
-

09

Module SQL

Course Module

Module 01

Introduction to MySQL

- Overview of relational databases and SQL
 - Understanding SQL syntax and commands
 - Overview of MySQL database management system
 - Installation and setup of MySQL server and client
-

Module 02

Understanding Basic SQL Queries and Functions

- Basics of the SELECT statement
 - Filtering data with WHERE clause
 - Sorting results with ORDER BY clause
 - Using SQL functions (e.g., COUNT, SUM, AVG)
 - Performing aggregate functions on data sets
-

Module 03

Advanced SQL Queries: Working with Tables and Subqueries

- Understanding table joins (INNER JOIN, LEFT JOIN, RIGHT JOIN)
 - Performing multi-table queries
 - Using subqueries to filter and manipulate data
 - Creating derived tables for complex queries
-

Module 04

Data Manipulation using SQL

- Inserting new records into tables
 - Updating existing records
 - Deleting records from tables
 - Understanding transactions and their properties
 - Implementing error handling in SQL scripts
-

Module 05

Database Administration and Optimization

- Managing databases, tables, and users in MySQL
- Backing up and restoring databases
- Indexing strategies for improving query performance
- Analyzing query execution plans
- Implementing security measures in MySQL

10

Module ARCGIS

Course Module

Module 01

Introduction to ArcGIS

- Overview of ArcGIS software and its components
 - Understanding GIS (Geographic Information Systems) and its applications
 - Installing Arc GIS software
 - Exploring the ArcGIS Desktop interface
-

Module 02

Understanding Data Types in ArcGIS

- Introduction to different types of GIS data (vector vs. raster)
 - Understanding shapefiles, geodatabases, and other GIS file formats
 - Creation of shapefiles (point, polygon and polyline)
-

Module 03

Introduction to Spatial Analysis Concepts and Geoprocessing Tools on ArcGIS

- Introduction to spatial analysis techniques and tools
 - Understanding spatial relationships and spatial queries
 - Overview of geoprocessing tools in ArcGIS
 - Performing common geoprocessing tasks (buffering, clipping, overlay analysis)
-

Module 04

Managing GIS Data on ArcGIS

- Creating new GIS datasets
 - Editing existing GIS data (attributes and geometry)
 - Creating and working with GIS data in ArcGIS
-

Module 05

Principles of Map Design and Cartographic Embellishments

- Understanding cartographic principles (layout, symbology, scale)
- Creating visually appealing maps in ArcGIS
- Using ArcGIS tools for thematic mapping
- Incorporating labels and annotations

Capstone Project

As part of the student evaluation all students will work on a personal research project in any topic of interest. Presentations of the research projects will mark the completion of the programme

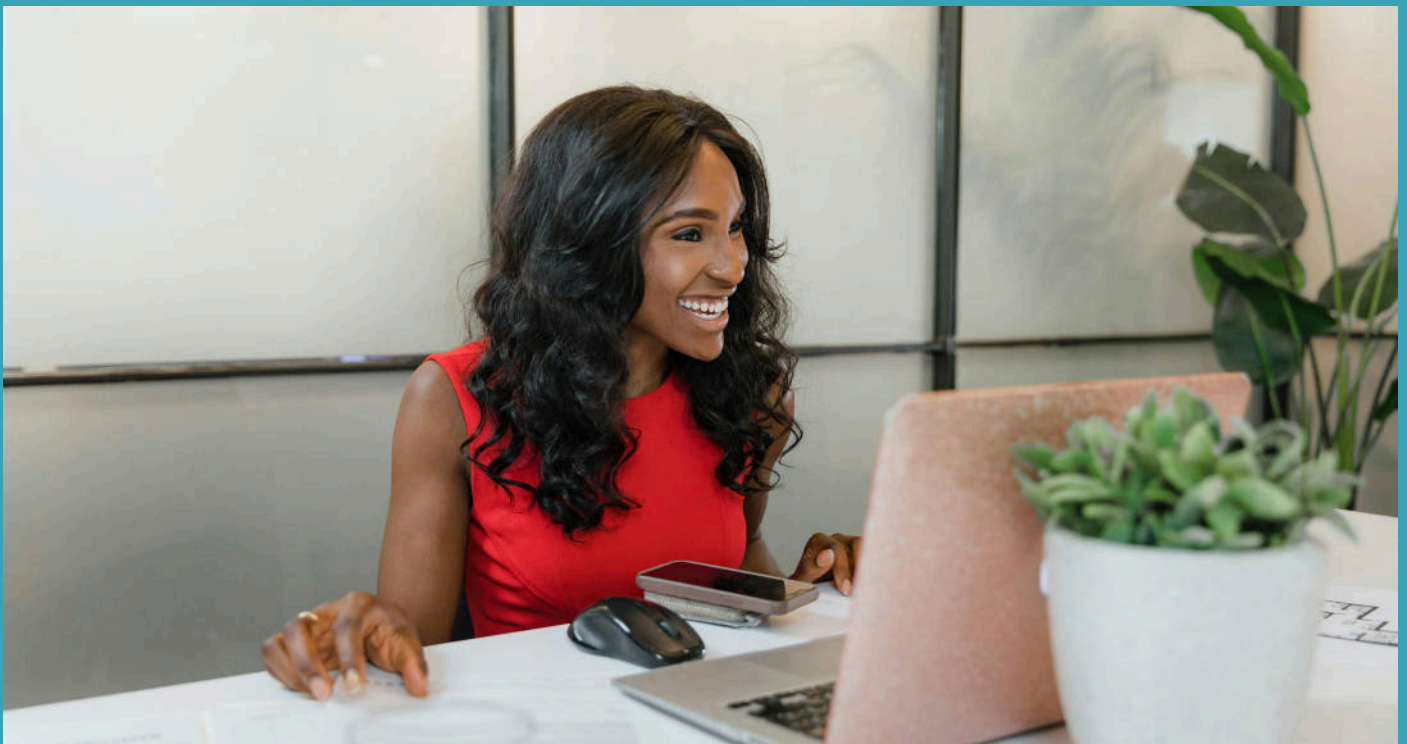
Laptop Specifications for the Bootcamp:

Os: Windows (preferably)

Ram: 4gb Minimum

Harddisk: 250gb Minimum

Processor: 2.0 Minimum (Dual core Preferably)



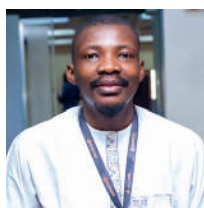


Contact:



Oche Loveth

Head of Operaitons,
Data-Lead Africa
+234-9030562587



Gabriel Mathew

Head of Communications
Data-Lead Africa
+234 - 0941631173



Mrs. Felicia Ayodele

Chief Operating Officer
Data-Lead Africa
+234 - 09166661234



Arowolo Ayoola (Ph.d)

Lead Partner
Data-Lead Africa
+234-07030300741



**“Upgrade your
Data Analytic Skills
and Accelerate
your Career!”**



Data-Lead Africa
Turning data to decision

10th street, Bassan Plaza, Block F, 3rd
floor, Central Business District, Abuja

    DATA-LEAD AFRICA